MOHAMED IBRAHIM

mohamed.abdelwahed.ai@gmail.com +201117599070 New Cairo, Egypt
linkedin.com/in/mohamed-ibrahim100 github.com/MohamedIbraheem100

PERSONAL PROFILE

AI Engineer focused on Agentic workflows and Applied Computer Vision. Proven capability in orchestrating multi-agent systems and deploying multimodal pipelines (LangGraph, SAM2, FastAPI) for low-latency production environments.

EXPERIENCE

AI Engineer – AL Master IT Solutions 04/2026 – Present
- Architect and deploy a production-grade, bilingual Agentic AI chatbot tailored for real-world customer interactions.
- Design scalable backend infrastructure leveraging FastAPI, Redis, PostgreSQL, and Vector Databases to support robust data workflows.
- Develop advanced Agentic RAG workflows using LangGraph, integrating secure authentication, rate limiting, and API defense mechanisms.
- Optimize inference and retrieval pipelines, successfully driving a 30%+ latency reduction to achieve sub-second response performance.
- Implement conversation persistence architecture and streamline CRM automation for enhanced data tracking.

AI Engineering Trainee – Advanced Computer Vision 01/2026 – 04/2026
National Telecommunication Institute (NTI)
- Designed and implemented end-to-end computer vision pipelines for classification, detection, and segmentation tasks.
- Improved and evaluated deep learning architectures including CNNs, Vision Transformers (ViT), and multimodal models.
- Built and optimized generative models (GANs, Autoencoders, Diffusion) for image synthesis and enhancement.
- Engineered LLM-driven workflows and AI agents using LangGraph for orchestration, task routing.
- Deployed interactive AI applications integrating model inference with user interfaces.

IT Engineer – Watanya Company For Roads (Military Service) 12/2024 – 11/2025
- Maintained and operated IP-based surveillance systems (NVRs, CCTV, RTSP streams) across multiple sites.
- Supported integration of face and license plate recognition systems with live camera feeds.
- Troubleshot system and network issues, improving uptime and reliability.
- Managed data storage and database operations for continuous monitoring systems.

EDUCATION

Bachelor's degree in Computers and Artificial Intelligence, 2020 – 2024
Fayoum University
Graduation Project : Cashy: AI Mobile App Currency Recognition & Counterfeit Detection | Grade : **Excellent**

PROJECTS

SpotLight — AI Advertisement Insertion System Capstone Project | NTI
Description: Built an end-to-end pipeline that inserts advertisements into existing video content (films, series, football matches).
Features:
- Integrated Qwen 2.5 VL 7B for scene understanding, OWLv2 for surface detection, and SAM2 for pixel-accurate foreground tracking across all frames.
- Implemented Perspective Homography (OpenCV) to warp ads to match surface angles, eliminating the floating ad problem.
- Applied Stable Diffusion Inpainting for surface-aware edge blending across walls, screens, grass, and tables.
- Optimized for limited GPU environments using 4-bit quantization and sequential VRAM management.

Cashy: AI Mobile App Currency Recognition & Counterfeit Detection (Graduation Project)
Description: Developed a multi-model AI system for real-time Egyptian currency identification and fraud detection.
Features:

- Dual-Model Architecture: Developed and trained two specialized YOLO models for currency classification and counterfeit detection, achieving a 95% mAP.
- Mobile Optimization: Converted deep learning models to TensorFlow Lite (TFLite) to ensure efficient, low-latency inference on mobile devices.
- System Integration: Built a Flask backend to bridge AI models with the mobile frontend.
- Pre-processing Pipeline: Implemented OpenCV Image preprocessing to enhance image quality and extract forensic features for high-accuracy banknote verification.

Storytelling Agent
Description: A multi-agent generative system that transforms user ideas into illustrated stories using an autonomous orchestration pipeline.
Features:
- Autonomous task routing between story writing and image prompt engineering agents.
- Context-Aware Prompts: Automated transformation of narrative themes into high-quality visual scene descriptors.
- Interactive multi-modal dashboard for seamless story and image rendering.

TECHNICAL SKILLS

AI & GenAI: LLMs, RAG, Agentic AI, Prompt Engineering, LangGraph, LangChain

Computer Vision: Object Detection, Segmentation, Vision Transformers, Multimodal AI, OpenCV, YOLO, SAM2

Deep Learning: PyTorch, TensorFlow, CNNs, GANs, Diffusion Models

Backend & Deployment: FastAPI, Flask, Docker, REST APIs

Databases: PostgreSQL, MongoDB, MySQL, Redis

CERTIFICATES

- National Telecommunication Institute (NTI) | Advanced Computer Vision Track.